Sybil-Robust Multi-Item Diffusion Auctions via Cluster Reduction

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Course Project

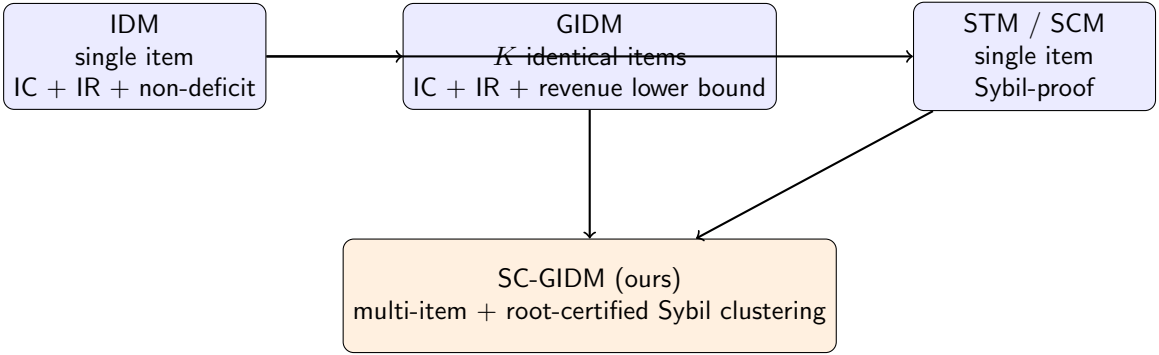
April 2026

Problem in one line

- IDM solves truthful *single-item* diffusion auctions.
- GIDM extends this to *multiple homogeneous items*.
- STM/SCM make the *single-item* setting Sybil-proof.
- **Open question:** how do we get Sybil-proofness for the multi-item case?

Our answer: in a restricted but rigorous setting, contract Sybil clusters to certified roots and then run GIDM on the quotient graph.

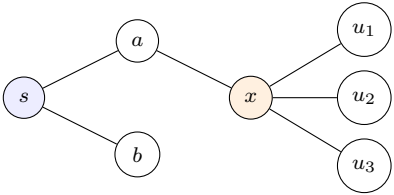
Flow from the baseline papers



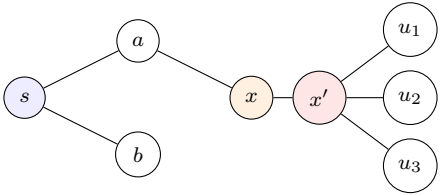
Project philosophy

Not "solve everything"; solve a clean restricted version with theorem-level guarantees and experiments.

Why naive multi-item diffusion is vulnerable



Honest graph



Chain Sybil attack

- Running GIDM directly on reported identities treats fake nodes as real bidders/intermediaries.
- Because GIDM reduces to IDM when $K = 1$, and IDM is known to be Sybil-vulnerable, identity-level GIDM cannot be Sybil-proof in general.

Our model and assumptions

Restricted setting

- K homogeneous items
- single-unit demand
- no collusion between distinct real buyers
- each Sybil cluster has a **certified non-Sybil root**

Why these assumptions?

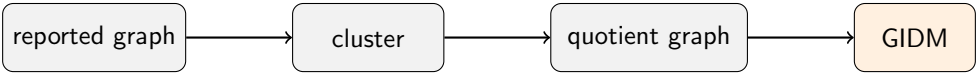
They are exactly what makes a clean reduction proof possible.
We do *not* claim a full detection algorithm for arbitrary Sybils.

Key design choice

The cluster bid is the **root bid**, not the maximum Sybil bid.

Mechanism: SC-GIDM

1. Partition reported identities into Sybil clusters.
2. Contract each cluster into one meta-buyer.
3. Use the certified root's bid as the cluster bid.
4. Merge all external edges into cluster-level diffusion edges.
5. Run baseline GIDM on the quotient graph.
6. If a cluster wins, allocate one item to its root and charge the root.



Main theorem idea

Lemma

After cluster contraction, any Sybil attack by a real buyer i becomes a feasible *single* meta-report

$$(\widehat{b}_i, \widehat{r}_i)$$

of the cluster root, with $\widehat{r}_i \subseteq r_i$.

Why?

- internal Sybil nodes disappear after contraction;
- external cluster edges can only go to true neighbours of the root;
- the mechanism ignores fake bids and keeps only the root bid.

Consequence

SC-GIDM is just GIDM on the quotient graph, so we inherit:

$$\text{IR} + \text{IC} + \text{weak budget balance.}$$

Sybil-proofness theorem

Theorem

Under homogeneous items, single-unit demand, no collusion, and certified roots, SC-GIDM is Sybil-proof against identity splitting.

Proof sketch.

1. Any Sybil attack collapses to a feasible deviation by one meta-buyer on the quotient graph.
2. GIDM is truthful on that quotient graph.
3. Cluster utility equals root utility because the cluster can receive at most one item.
4. So creating fake identities cannot improve the real buyer's utility.

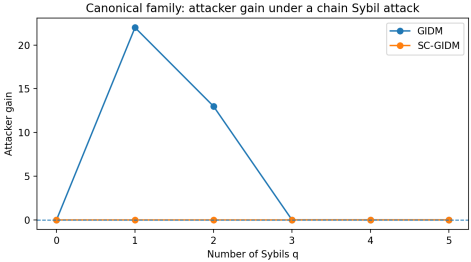
□

Canonical profitable attack

- $K = 3$
- honest GIDM winners: b_1, b_2, b_7
- attacker b_0 utility: 0
- attack: add one fake identity b_0' with the same bid 98 and place it between b_0 and its neighbours

$$p_{b_0'} = 76, \qquad p_{b_2} = -75$$

$$u_{b_0}^{\text{attack}} = 98 - 76 = 22$$



What SC-GIDM does

Cluster $\{b_0, b_0'\}$ is contracted back to root b_0, the fake bid is ignored, and the attack gain returns to 0.

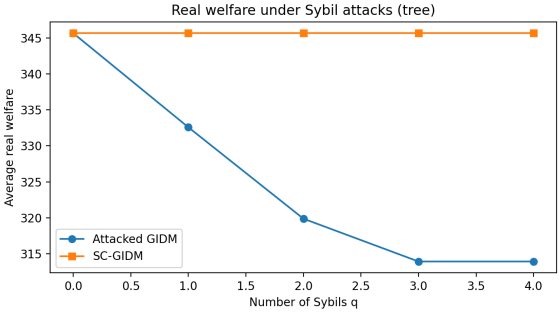
Simulation setup

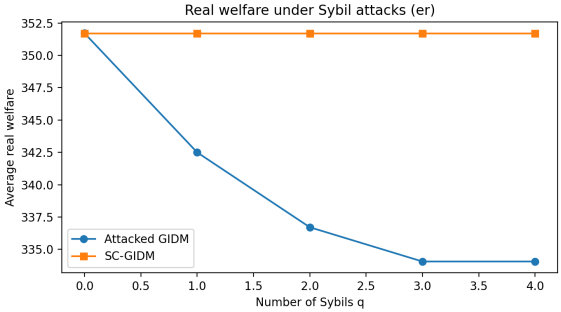
- Graphs: random trees and connected Erdős-Rényi graphs
- $n = 20$ real buyers, $K = 4$, 3 seller neighbours
- attacker = highest betweenness / degree buyer
- attack = chain Sybil attack with $q \in \{0, 1, 2, 3, 4\}$
- 120 random seeds

Compared mechanisms

1. Local-K-Vickrey
2. honest GIDM baseline
3. attacked GIDM
4. SC-GIDM

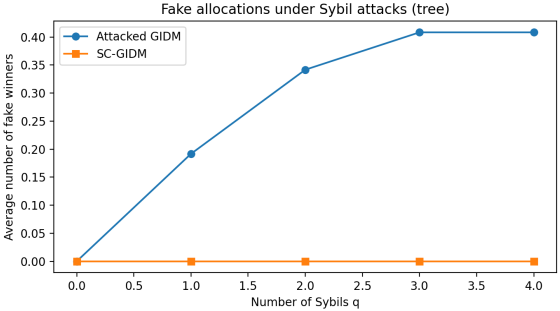
Result 1: real welfare is recovered

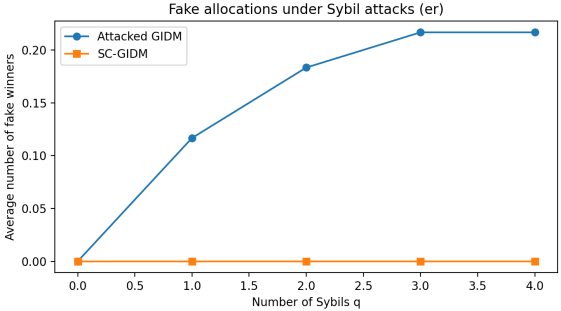




- Attacked GIDM loses real welfare as q grows.
- SC-GIDM stays essentially on the no-attack curve.
- So the cluster reduction restores *real* allocation quality.

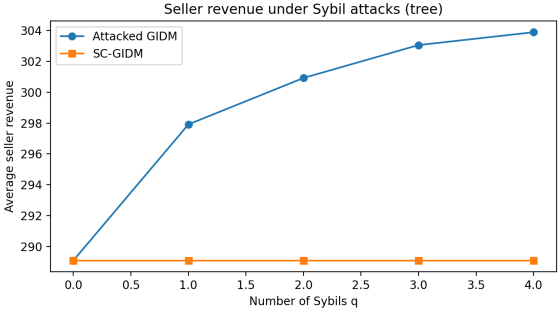
Result 2: fake winners disappear

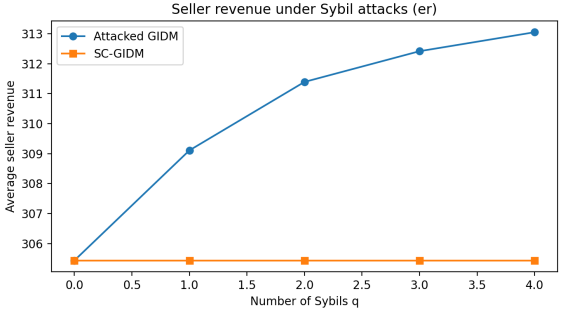




- Ordinary attacked GIDM increasingly allocates items to fake identities.
- SC-GIDM eliminates fake winners because it contracts them before the auction runs.

Revenue is not the whole story





- Fake identities can create extra competition, so attacked GIDM may show slightly higher *nominal* revenue.
- But this comes with fake winners and lower real welfare.
- Therefore robustness should be evaluated by real-buyer welfare and attack-resistance, not just seller revenue.

What we proved vs. what remains open

Proved

- quotient-graph reduction
- IR
- cluster-level IC
- weak budget balance
- Sybil-proofness against identity splitting

Still open

- imperfect / heuristic Sybil detection
- collusion across real buyers
- multi-demand and heterogeneous items
- multiple sellers

Takeaway

Main message

A clean way to extend the single-item Sybil-proof direction to multiple items is:

$$\text{certified Sybil clustering} + \text{GIDM on the quotient graph.}$$

- The mechanism is rigorous under a clear restricted model.
- It answers an explicit open question from the Sybil-proof paper.
- It works well experimentally on the metrics that matter for robustness.

Thank you